Thomas O'Neill

Systems & AI Engineer

10 years of full-stack and infrastructure

Selected Projects, 2026

Upfront

Oct 2026 – Present

Rust Wayland shell built as an operating system: work sits around you, comes close or recedes, and a Claude Code agent you talk to keeps the room in order. ~50k lines, written and graded by agents.

Claude Code · Loop engineering · Skills & hooks · Subagents · TypeSafe · Rust · wgpu

– Loop engineering: each change ships a demo film; a script and blind agent judges grade it, and their verdicts write the next brief.

– Built on three-rs, my first loop-engineered project: Three's own example screenshots were the definition of done; 100/215 pass.

d3-three `github.com/oneilltomhq/d3-three`

Oct 2026

18 D3 gallery charts in three.js WebGPU, built with Claude Code. Live demo on GitHub Pages.

Claude Code · Reference-graded evals · Three.js · D3

– Each page rendered headless, diffed against a committed frame and checked against its data.

semantic-atlas `github.com/oneilltomhq/semantic-atlas`

Feb 2026

Your ChatGPT or Claude history as a 3D point cloud: embeddings, UMAP, HDBSCAN, then WebGPU.

Python · Embeddings · Three.js · WebGPU

– Python pipeline (sentence-transformers, BERTopic) and a Three.js WebGPU explorer; local-only.

agent-browser-ext `github.com/oneilltomhq/agent-browser-ext`

Feb 2026

Chrome extension prototypes: CDP automation primitives for in-browser agents; voice-browser-agent adds a voice side panel.

TypeScript · CDP · Deepgram

Work Experience

AI Engineer @ Independent

Mar 2025 – Present

Self-funded, full-time on agentic engineering with Claude Code; since Oct 2026 building Upfront.

– Briefs, evals and guardrails instead of hand-written code; smallest model per step; parallel subagents.

Lead 3D Web Engineer @ Open Systems Lab

Nov 2021 – Mar 2025

Led frontend and 3D on BuildX, an open-source housing configurator; EU pilot, then WikiHouse.

– Production 3D design tool in React, Next.js, Three.js; BIM pipelines (Speckle, IFC, BOM/OBJ/GLB).

Co-Founder & Lead Developer @ Bearjam

Jan 2020 – Nov 2021

Small web studio: client software and an Arts Council England funded GPU image editor.

Systems Engineer & Developer @ Claranet

Nov 2015 – Apr 2017

Infrastructure automation (Docker, Kubernetes, Ansible); internal web UI and REST API.

Contact

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London / remote

Contract · outside IR35

Skills

Agents & LLMs

Claude Code · Claude Agent SDK · Prompt & instruction design · Skills & hooks · Subagents · Loop engineering · Reference-graded evals · Guardrails · Structured output · Voice (STT/TTS)

Engineering

TypeScript · Python · Rust · React / Next.js · Cloudflare Workers · Playwright · Postgres / SQLite · Three.js · WebGPU · Docker · Kubernetes · Linux